

contract — Customer contract setting

A contract is a method of paying for phone numbers that MewbileTech offers customers. The company offers three contracts to pay for a phone number: namely through a Term contract, a Month-to-Month contract, and a Prepaid contract. These contracts cycle monthly, and customers are billed according to their selected contracts.

A Term contract sets a time frame in which a customer will use the contract for. This results in lower monthly fees, lower cost per minute of calling, and free minutes of calling, but requires the customer to give a deposit beforehand as collateral in case the customer cancels the contract before the time frame ends.

A MTM contract does not have a set time frame which requires customers to fulfill. Customers can cancel their contract at any time, but instead they have higher monthly fees, higher cost per minute of calling, and no free minutes, unlike that of a Term contract.

A Prepaid contract uses a credit system where the customer must deposit money into their contract. This contract has the lowest cost per minute of calling and does not require the customer to fulfill any agreements. Cancellation can occur at any time, but if the credit is positive, the customer must pay the company the corresponding amount.

The contract module implements classes to define the type of phone plan a customer needs for a specific phone number. As a result, this module allows other employees to quickly add a contract to a specific customer's phone number without needing to know the details of the contract itself.

Module Contents

The contract module defines the following classes:

class Contract.**TermContract**(start, end, bill)

Creates a specialized contract object which functions with a starting and ending date. The mandatory start parameter is what sets the beginning date of the term contract which consists of a month and year. Similarly, end is another mandatory parameter which sets the end date of the term contract, also consisting of a month and year. These two parameters define the scope of the bill parameter which is a list of Bill objects.

The TermContract class provides 3 methods:

new_month(month, year, bill)

Cycles a customer's bill to the next corresponding month and year by using the month and year parameters. By incrementing to a new date, it is expected that the customer has paid off the bill of the previous month and all arguments are reset to their defaults specified by a term contract. Moreover, this method compares whether the end date is less than the month and year parameter which will indicate to a private attribute whether the customer may receive their term deposit back.

This method will also be used to start the first month of new customer term contracts.

bill_call(call)

This method uses a call parameter based on Call object that will calculate the duration of all calls, track number of free minutes used, and add to bill all additional minutes after the given free minutes have been used.

It is expected that a bill has already been created for the current month and year when the Call objects are made and that the bill parameter is in the correct month and year.

cancel_contract

When the customer wishes to cancel a term contract, return the amount that the customer owes in the cancellation month. Similarly, this method will check whether the customer is eligible for their deposit through a private attribute.

It is expected that a bill exists for the month and year of cancellation.

class Contract.MTMContract(start, bill)

Creates a specialized contract that functions with only a start date. The mandatory start parameter sets the first month and year of the contract for the customer. Correspondingly, the bill will follow and start on the start date until the customer cancels the contract.

The MTMContract class provides 1 method:

new_month(month, year, bill)

Cycles a customer's bill to the next corresponding month and year by using the month and year parameters. By incrementing to a new date, it is expected that the customer has paid the bill of the previous month and all arguments are reset to their defaults specified by a MTM contract.

This method will also be used to start the first month of new customer MTM contracts.

class Contract.**PrepaidContract**(start, bill, balance)

Creates a specialized contract that functions with a start date. The mandatory start parameter sets the first month and year of the contract for the customer. Correspondingly, the bill will start on this start parameter and follow until the customer cancels the contract. Additionally, the customer must also provide any amount of money to their contract for their use.

It is expected that the customer cannot provide negative amounts of money.

The PrepaidContract provides 2 methods:

new_month(month, year, bill)

Cycles a customer's bill parameter to the next corresponding month and year by incrementing the month and year parameters. It is not expected for a customer to pay off any debt they have accrued when advancing to a new month in this contract, however if the balance falls below 10 dollars, 25 dollars will automatically be added to the customer's prepaid contract.

This method will also be used to start the first month of new customer Prepaid contracts.

cancel_contract

When the customer wishes to cancel a prepaid contract, the method will check whether the contract's balance is greater than 0, if so, the balance will be added to the amount the customer needs to pay before the cancellation is finalized. If the customer still has credit at the cancellation, the remaining amount is forfeit and will not be refunded.

It is expected that a bill exists for the month and year of cancellation.

The contract module defines the following constants:

contract.**MTM_MONTHLY_FEE**

Instructs MTMContract objects to add the monthly fee for a MTM contract to a customer's monthly bill.

contract.**TERM_MONTHLY_FEE**

Instructs TermContract objects to add the monthly fee for a Term contract to a customer's monthly bill.

contract.**TERM_DEPOSIT**

Instructs TermContract objects to add the fee that a customer must pay the company as collateral in case the customer starts and cancels a Term contract before the agreed cancellation month.

contract.**TERM_MINS**

Instructs TermContract objects to add a set number of free minutes customers may use to call before customers must pay for them to a customer's monthly bill.

contract.**MTM_MINS_COST**

Instructs MTMContract objects to set the amount of money that a customer must pay per minute of calling to a customer's monthly bill.

contract.**TERM_MINS_COST**

Instructs TermContract objects to set the amount of money that a customer must pay per minute of calling to a customer's monthly bill.

contract.**PREPAID_MINS_COST**

Instructs PrepaidContract objects to set the amount of money that a customer must pay per minute of calling to a customer's monthly bill.